Independent replication of “Implementation of the Quantum Fourier Transform” (Weinstein, Lloyd, Cory, quant-ph/9906059)

QC-200 wave, replication host = CherryRd

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Abstract

We independently replicate the algorithmic content of Weinstein, Lloyd, and Cory’s 1999 paper on implementing the Quantum Fourier Transform (QFT) on a 3-qubit NMR quantum computer. The paper’s testable core is the Coppersmith decomposition of the QFT into n Hadamards and $n(n-1)/2$ controlled-phase gates, and the explicit QFT_4 matrix given in its Eq. 4. Using Qiskit 2.5.0 we build the Coppersmith circuit for $n = 3, 4, 5$ and verify (i) the gate count is exact ($H = n$, $\mathrm{CP} = n(n-1)/2$), (ii) the output state matches the analytic QFT amplitudes on every computational-basis input to $\leq 5.6 \times 10^{-15}$, (iii) the $n = 2$ operator matches paper Eq. 4 to 1.15×10^{-16}, and (iv) our circuit agrees with `qiskit.circuit.library.QFT` to zero after removing a global phase. The paper’s NMR hardware fidelity numbers ($F = 87\%$ overall, $> 98\%$ per gate) are hardware claims and are not reproducible on an ideal statevector simulator; we note the correct interpretation and scope. **Verdict: REPLICATED** for the paper’s algorithmic and computational claims; the hardware claims are marked *OUT-OF-SCOPE-BY-DESIGN.*

1 Paper summary

Weinstein, Lloyd, and Cory (MIT, 1999; arXiv:quant-ph/9906059v1) implement the QFT on the three C-13 spins of an alanine molecule at 9.4 T on a liquid-state NMR spectrometer. The QFT is defined (paper Eq. 1) as

$$\mathrm{QFT}_q\,|x\rangle \to \frac{1}{\sqrt{q}}\sum_{p=0}^{q-1} e^{2\pi i a p/q}\,|p\rangle,$$

and for the two-qubit ($q = 4$) case reduces to the explicit unitary in paper Eq. 4. The paper adopts the Coppersmith decomposition of the QFT (paper Eqs. 5–7): A_j is a Hadamard on qubit j, and B_{jk} is a controlled-phase gate with angle $\theta_{jk} = \pi/2^{k-j}$ on the (j, k) pair. The composite gate $B_{j,j+1}B_{j,j+2}\ldots B_{j,L-1}A_j$ is applied on the “lead bit” $j = L-1$, then the sequence is repeated with j decremented to 0; this gives an $O(n^2)$-two-qubit-gate QFT. The paper then translates each Coppersmith gate into an NMR pulse sequence for a J-coupled 3-spin system (paper Eqs. 9–11), and reports:

- overall NMR fidelity $F = 87\%$ using the Uhlmann-like measure in paper Eq. 12;
- after renormalizing ρ_{exp} to compensate for decoherence attenuation, per-gate fidelity $> 98\%$ over the six gates of the 3-qubit QFT.

2 Claims table

#	Claim	Type	Testable in ideal sim?	Tested here?
C1	QFT decomposes into $A_j = H$ (paper Eq. 5) and $B_{jk} = \mathrm{CP}(\pi/2^{k-j})$ (paper Eq. 6).	algorithmic	YES	YES
C2	For an L-qubit register the Coppersmith QFT uses n Hadamards and $n(n-1)/2$ controlled-phase gates ($O(n^2)$ total two-qubit gates).	complexity	YES	YES (n=3,4,5)
C3	The QFT satisfies paper Eq. 1 for all computational-basis inputs.	correctness	YES	YES (n=3,4,5)
C4	QFT_4 equals the explicit matrix in paper Eq. 4.	correctness	YES	YES
C5	Alanine C-13 3-spin NMR realization achieves overall $F = 87\%$.	hardware	NO	OUT-OF-SCOPE (needs a real spectrometer)
C6	After renormalizing ρ_{exp} to compensate for decoherence, per-gate NMR fidelity is $> 98\%$ over 6 gates.	hardware	NO	OUT-OF-SCOPE

3 Method

3.1 Environment

- Host: `CherryRd` (macOS; x86_64), Python 3.14.6.
- Qiskit 2.5.0, NumPy 2.5.1, MarkItDown for the marker-side extraction (see failure analysis for why Marker/Nougat proper were not used).
- All simulation is exact statevector; no NMR / density-matrix / open-system simulation.

3.2 Circuit construction (Coppersmith)

For $j = n-1, n-2, \ldots, 0$ apply

$$H_j,\ \mathrm{CP}(\pi/2^{j-k})_{k\to j} \quad \text{for } k = j-1, \ldots, 0.$$

Optionally append the bit-reversal SWAP. This is a direct transcription of paper Eq. 7 into Qiskit's little-endian convention. See `report/evidence/qft_replication.py`, function `coppersmith_qft`.

3.3 Verification procedures

1. **Gate count** (`verify_gate_count_claim`): build the circuit without the trailing SWAP, count `h` and `cp` operations, compare to n and $n(n-1)/2$.

2. **Analytic correctness** (`verify_correctness`): for every $x \in \{0, \ldots, 2^n - 1\}$ prepare $|x\rangle$, apply the circuit, and compare the resulting statevector to $\frac{1}{\sqrt{2^n}} \sum_y e^{2\pi i xy/2^n} |y\rangle$.
3. **Explicit matrix** (`verify_paper_eq4_matrix`): build the full 4x4 operator for $n = 2$ and compare to paper Eq. 4 entry-by-entry.
4. **Sanity against Qiskit's own QFT** (`compare_to_qiskit_builtin`): fold out a possible global phase between our operator and `qiskit.circuit.library.QFT`.
5. **Anchor cases** (`anchor_test_case`): print amplitudes for $x = 0, 1, 3, 7$ at $n = 3$.

Reproduce with:

```
cd QC-quant-ph-9906059-quantum-fourier-transform-implementation
python3 -m venv .venv && source .venv/bin/activate
pip install qiskit numpy markitdown
python report/evidence/qft_replication.py
```

4 Results vs. paper

4.1 Gate count (C2)

n	H measured	H expected	CP measured	CP expected	match?
3	3	3	3	3	YES
4	4	4	6	6	YES
5	5	5	10	10	YES

Formula $n(n-1)/2$ for controlled-phase count is exactly satisfied.

4.2 Analytic correctness (C3)

Max ℓ_∞ error over all 2^n computational-basis inputs:

n	inputs checked	max amplitude error vs analytic	to machine precision?
3	8	1.42×10^{-15}	YES
4	16	3.78×10^{-15}	YES
5	32	5.62×10^{-15}	YES

4.3 Paper Eq. 4 matrix (C4)

Our QFT_4 operator matches the explicit matrix in paper Eq. 4 to 1.15×10^{-16} in every entry.

4.4 Sanity vs Qiskit built-in QFT

For each $n \in \{3, 4, 5\}$, our Coppersmith operator equals `qiskit.circuit.library.QFT` to floating-point zero after absorbing a global phase; phase variation across nonzero entries is zero to floating-point.

4.5 3-qubit anchor cases

For $x = 0, 1, 3, 7$ at $n = 3$: max amplitude error vs $\frac{1}{\sqrt{8}} \sum_y e^{2\pi i x y/8} |y\rangle$ is $\leq 5 \times 10^{-16}$ in every case (see `report/evidence/results.json`, key `anchor_3qubit_selected_inputs`).

4.6 Hardware claims (C5, C6)

Not tested. These are NMR spectrometer measurements on an alanine sample; they require an actual 9.4 T NMR magnet and a J-coupled ^{13}C sample. We note that the paper's per-gate 98% comes from a renormalization argument (isotropic attenuation of the deviation density matrix) that is not obviously exact when J_{13} is 30–45x smaller than J_{12}/J_{23}, so refocusing intervals approach T_2; we flag this in Open Question Q2.

5 Verdict

REPLICATED for the paper's algorithmic and computational content:

- C1 (Coppersmith decomposition): reproduced exactly.
- C2 (gate count $n(n-1)/2$ CP + n H): reproduced exactly for $n = 3, 4, 5$.
- C3 (analytic correctness on all basis inputs): reproduced to machine precision.
- C4 (paper's Eq. 4 QFT_4 matrix): reproduced to 10^{-16}.

Hardware claims C5 and C6 are *OUT-OF-SCOPE-BY-DESIGN* for a statevector replication and are not evaluated as MATCH/MISMATCH.

6 Open Questions

Q1. How well does the Coppersmith QFT circuit, dropped into the paper's Eq. 11 pulse program, reproduce the reported 87% overall NMR fidelity when composed with a realistic amplitude-damping + phase-damping channel calibrated to $T_1 = 1.56\,\text{s}$ and $T_2 = 420\,\text{ms}$ on alanine C-13 at 100.6 MHz, rather than an isotropic depolarizing channel? (Motivated by: our ideal statevector rules out the algorithmic decomposition as a source of the 13-point fidelity gap, but we did not run an open-system simulation.)

Q2. The paper's "renormalize ρ_{exp} to lift F from 87% to 98%/gate" step assumes decoherence acts as an isotropic scalar attenuation on the deviation density matrix. Under which noise channels is that actually valid, and does it hold when $J_{13} = 1.2\,\text{Hz}$ forces $1/(8J_{13}) \sim 100\,\text{ms}$ refocusing intervals comparable to T_2?

Q3. The paper avoids an explicit SWAP by "reordering the bits at the appropriate interval". The exact mapping from (carbonyl, C_α, C_β) to Qiskit qubit indices is not printed, blocking a bit-for-bit reproduction of Fig. 1's A/B/C spectrum-to-qubit assignment.

Q4. Up to what n is the Coppersmith $n(n-1)/2$ controlled- phase count the actual optimum for the QFT on an all-to-all register, before approximate-QFT truncation of small-angle B-gates or restricted connectivity change the optimum?

Q5. Given our exact ideal-statevector result at $n = 3, 4, 5$, on modern superconducting/trapped-ion hardware with two-qubit gate error $\sim 10^{-3}$, what is the crossover $n^{\star}$ at which the full Coppersmith QFT becomes indistinguishable, via Eq. 12's F, from a Haar-random unitary?

See `report/open_questions.json` for the machine-readable version with `next_steps`.

7 Artifacts

See `report/artifacts_summary.md` for the full 8-artifact inventory and `report/failure_analysis.md` for the residual gaps (Marker/Nougat model weights not on this host; hardware claims by-design out of scope).